COMP3821 Project

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Contents

Tags: Data Structures & Algorithms

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1.1 Brainstorming

1.1.1 Idea 1: Digital circuit optimisation

1. Papers: `https://arxiv.org/abs/2509.13966`

1.1.2 Idea 2: String edit distance

1. Papers: `https://dl.acm.org/doi/pdf/10.1145/3564246.3585178`

1.1.3 Idea 3: Markov chain mixing

Baskically, given a markov chain that can settle to a stationary distributioon, take enough samples such that it will settle to that distribution within some margin of error.

1. Possible Problems Find a faster algorithm for markov chain mixing (spoiler: we will fail). Idk maybe make it less precise or some shit. Narrow the specification for the type of Markov chain (i.e. find some condition that makes it settle faster / slower) and design an algorithm for

that. My idea: An algorithm for mixing discrete time-inhomogeneous Markov chains. Prove equivalence with some other problem and use it to approximate that.

2. Papers

   https://arxiv.org/pdf/2504.02740

   (a) https://web.archive.org/web/20170922034813/https://people.eecs.berkeley.edu/~sinclair/perm.pdf « this one is like "the paper" in the field. $1+\varepsilon$ approximation means for nonnegative real numbers $a, \tilde{a}, \varepsilon,\ a(1+\varepsilon)^{-1} \leq \tilde{a} \leq a(1+\varepsilon)$. Overall approach: "Given a graph G, construct a Markov chain whose states correspond to perfect and "near-perfect" matchings in G and which converges to a stationary distribution that is uniform over the states. Transitions in the chain correspond to simple local perturba- tions of the structures. Then, provided convergence is fast enough, we can generate matchings almost uniformly by simulating the chain for a small number of steps and outputting the structure corresponding to the final state."

      i. Referenced papers https://projecteuclid.org/journals/communications-in-mathematical-physics/volume-25/issue-3/Theory-of-monomer-dimer-systems/cmp/1103857921.pdf J. KEILSON, Markov Chain Models–Rarity and Exponentiality,

3. Concepts

   (a) Permanent Just the determinant but with no signs on the terms. Only really useful for combinatorics etc.

       These things are fucked up to try to compute naively.

   (b) Cycle cover Permanents can be used to find cycle cover of a graph. Each vertex $i \in V$ in the digraph has a unique successor $\sigma(i)$ on the cycle cover, and thus σ represents a permutation on V. Thus,

$$\text{weight}(\sigma) = \prod_{i=1}^{n} a_{i,\sigma(i)}$$

$$\text{perm(A)} = \sum_{\sigma} \text{weight}(\sigma)$$

       or, the permanent of A is equal to the sum of the weights of all cycle-covers of the digraph.

(c) Perfect matching If you view a square matrix A as the adjacency matrix of a bipartite graph, then for a perfect matching σ,

$$\text{weight}(\sigma) = \prod_{i=1}^{n} a_{i,\sigma(i)}$$

where $a_{i,j}$ is the weight of the edge.

(d) Things in the field

i. Van de Waerden's conjecture (Proven)